CATERPILLAR®

BEST PRACTICE SERIES

Managing Fuel Consumption Reduces Cost

Application **Maintenance** Site Management Component Rebuild Component Life Management Safety MARC Management

June 2011
0808-2.15-1129

300 Hamilton Blvd., Ste. 300, Peoria, IL 61629-3810, U.S.A.

1.0 Introduction

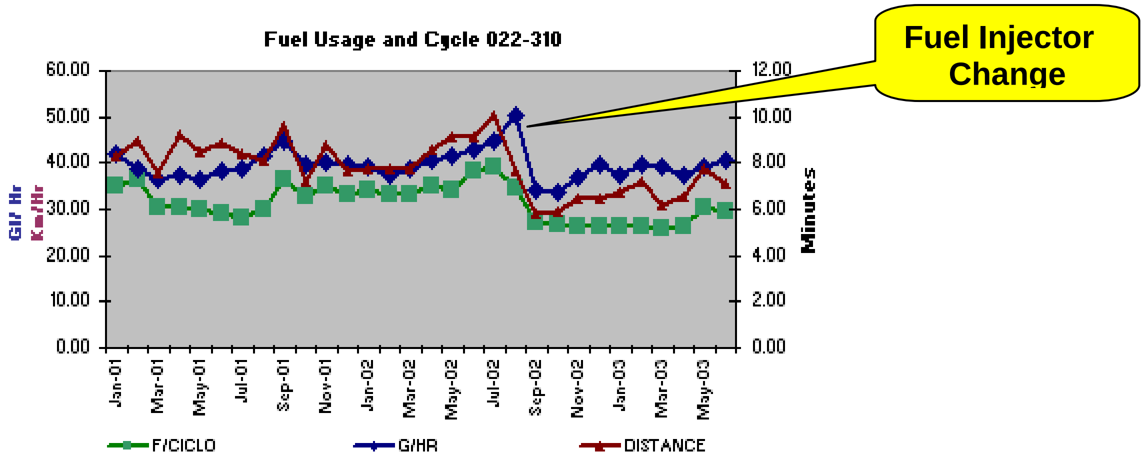
Fuel consumption has become one of the highest costs of owning and operating a machine. Monitoring fuel burn rates and performing a mid-life fuel injector change can save thousands of gallons of fuel.
This Best Practice discusses a fuel consumption monitoring process from a coalmine using 793C machines and illustrates the benefits of changing fuel injectors at engine mid-life.

2.0 Best Practice Description

As fuel injectors wear, fuel efficiency is affected and increase in average fuel consumption occurs. The leading cause of fuel injector wear is dirty fuel (water and dirt). There are several published Best Practices at this website that discuss fuel cleanliness and ways to achieve improved cleanliness. Clean fuel extends injector life and performance.
However, not every mine operation is able to maintain a consistent or acceptable fuel cleanliness level to prevent premature injector wear. An option for this situation is to monitor the machine rate of fuel consumption on a monthly basis, to change fuel injectors if a growth in fuel consumption is observed, and then to monitor the change.
It is not necessary to monitor every machine in a fleet. Begin by monitoring a few selected machines with a range of different hours on the engines. This monitoring will provide data on fuel efficiency and any changes over time. Note: If there is a fuel cleanliness issue, it will likely affect all of the machines in operation.
It is also necessary to monitor distance traveled. Distance traveled data is needed to determine if a change in fuel consumption rate is due to an operational change or an engine efficiency change.
Fuel consumption, distance, and cycle time data can be downloaded from the machine VIMS Payload Summary Report.

3.0 Implementation Steps

The graph below is data from a coalmine 793C. In this study, rate of fuel burn, distance traveled, and load cycles times were tracked monthly. The trend was then graphed.
This study is several years old, but with the increase in fuel costs over the past few years, the benefit of managing fuel consumption today has far more importance and impact.



Data Table

	May-01	Jun-01	Jul-01	Aug-01	Sep-01	Oct-01	Nov-01	Dec-01	Jan-02	Feb-02	Mar-02	Apr-02	May-02	Jun-02	Jul-02	Aug-02	Sep-02
	36.44	38.23	38.78	41.23	43.41	39.79	40.30	39.66	39.29	37.26	38.99	40.42	41.44	43.41	44.64	50.45	34.28
smu		32114	33144.1	34137.5	34651.7	35181.5	35707.7	35707.7	36240.9	36781.1	37308	37809	38312	39302.2	39302.9	39795.4	40299.2
RPM		1755	1773	1769	1738.5	1625.5	1620.5	1620.5	1732.5	1658	1736	1725	1735	1705	1741	16.98	1748.5
Presión Oil		64.16	69.23	68.35	61.84	64.95	64.16	64.16	65.76	64.85	66.56	68.02	61.34	69.08	69.67	69.02	66.12
H2O		0.65	0.79	0.25	0.78	1	1	1	0.31	0.34	0.63	0.25	0.88	0.84	0.81	0.63	0.25

Note that beginning fuel consumption averaged 39 – 43 gallons per hour. Growth from June to August took fuel consumption up to 50 gallons per hour with little change in haul distances. Injectors were changed in September, and fuel consumption dropped to an average 40 gallons per hour. This represents a 10-gallon per hour savings. When a 3500 engine starts burning more fuel, a noticeable increase in exhaust temperatures will occur. Increased exhaust temperatures may cause engine de-rate events, which can also lead to premature exhaust valve failures. The engine monitored had close to 8000 hours when the fuel injectors were changed.

Assuming that the new fuel injectors performed the same as the original injectors, the savings over the next 8000 hours (engine rebuild PCR is 16,000 hours) would average 5 gallons an hour for a total 40,000 gallons. This can also reduce exhaust temperatures and prevent premature valve failures.

Steps to replicate this project are:

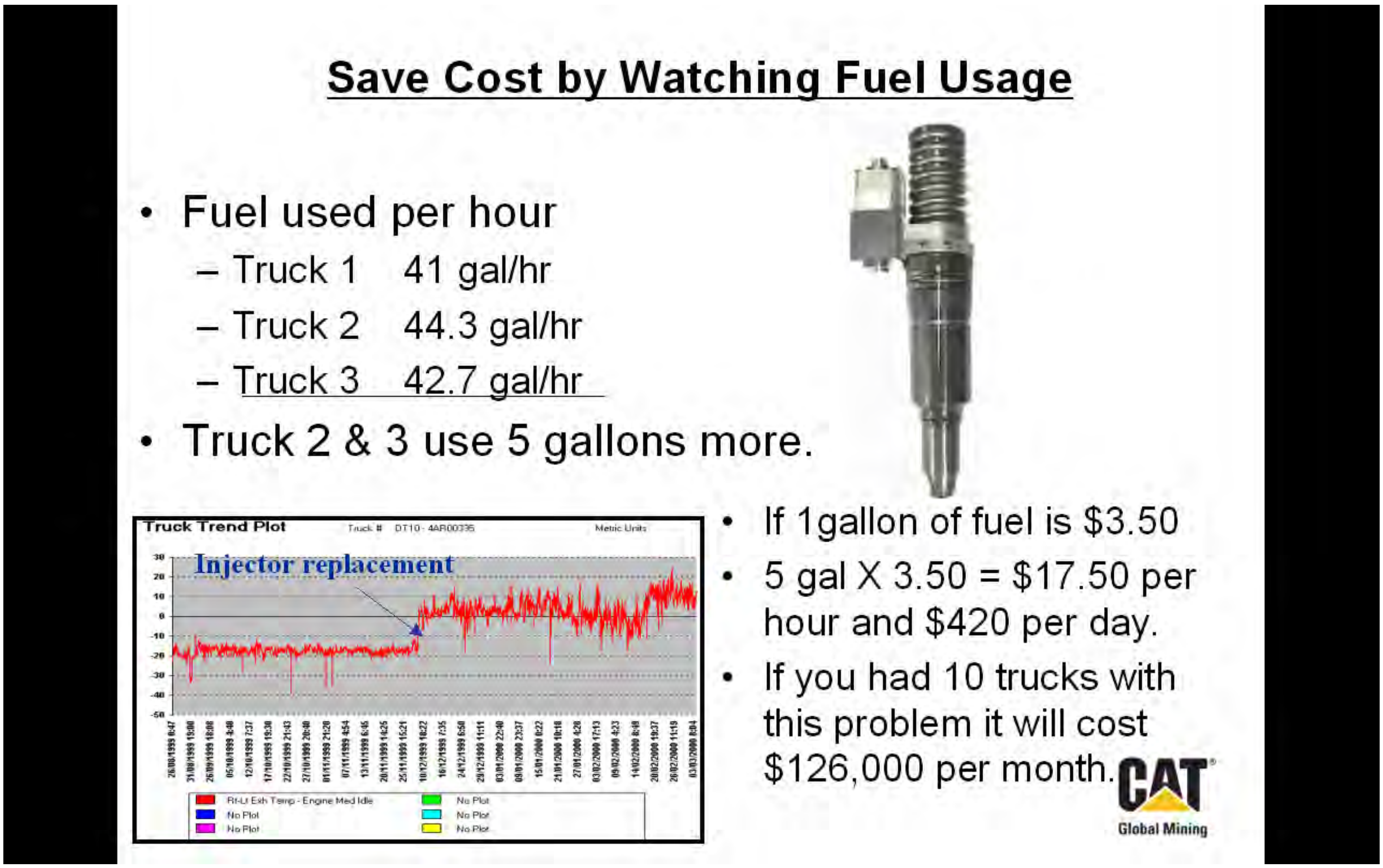
Simple test:

- Select an OHT with an engine that has approximately 8000 hours (mid-life). Measure fuel burn rate and distance traveled for thirty days and compare it to other trucks in the fleet. If the test truck is burning more fuel than the fleet average, then change injectors or investigate turbocharger operation which may be causing low boost. Monitor change in fuel burn.

Extensive test – validation.

- Select a fleet of machines for monitoring.
- Track fuel burn rate and distance traveled (monthly) for each machine and graph results.
- Identify any growth in fuel burn rate or difference in fuel burn rates between machines.

- If an engine shows higher than average fuel consumption, check boost pressure and injectors. If no mechanical problem is found, change the injectors and monitor fuel consumption results.
- Determine an "Average Injector Efficient Life"
- Measure cost (approx $19,000) to change injectors versus fuel savings.
- Determine a standard Preventative Maintenance policy for fuel injector replacement.



Note: Caterpillar has always recommended an injector and turbocharger change at engine mid-life. This recommendation is largely based on poor average fuel cleanliness at many locations around the world. Locations that have excellent fuel cleanliness typically would see little change in fuel injector efficiency over the full life of the engine.

4.0 Benefits

- 10% plus fuel savings

5.0 Resources Required

- Process to collect and analyze data.
- Monitor VIMS Payload Summary Reports.

6.0 Supporting Attachments / References

None at this time.

7.0 Related Best Practices

See Best Practices listed under "Component Life Management" category:

- o "Bulk Fuel Storage and Handling"
- o "Bulk Fuel Filtration"
- o "Series High Efficiency Filters"

8.0 Acknowledgements

This Best Practice was written by:
Bill Hasty
Product Support Consultant
Condition Monitoring
HASTY_BILL_T@cat.com

James McCarty
Product Support Consultant
Maintenance and Repair
Mccarty_James_T@cat.com

Dale Brehm
Six Sigma Black Belt
Global Mining Product Support
brehm_dale_e@cat.com